

Comprehensive Energy Consumption Analysis and Prediction using Machine Learning and LSTM

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Abstract—Forecasting energy use is an important part of sustainable development because it helps people use resources more efficiently, manage demand, and make better decisions. However, accurately modeling energy consumption remains challenging because it changes over time, has seasonal effects, and is affected by things outside of the home.

This study offers an extensive analysis and a predictive modeling framework utilizing electricity and fuel consumption datasets. The electricity dataset goes from 2013 to 2024 and has daily data points, which lets you look at short-term trends in detail. The fuel dataset goes from 1998 to 2025 and has monthly data points, which lets you look at long-term trends in energy use.

A thorough exploratory data analysis (EDA) is done to find hidden patterns in the data, such as seasonal changes, spikes in consumption around holidays, and differences in energy use between states. These analyses enhance comprehension of consumption behavior and facilitate proficient feature engineering.

We use performance metrics like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2) to test and compare different machine learning models, such as Linear Regression, Decision Tree, and Random Forest. Additionally, a Long Short-Term Memory (LSTM) network is employed to find patterns in the time-series data that depend on time. The outcomes indicate that Linear Regression performs better because the dataset is mostly linear. On the other hand, the LSTM model does a good job of capturing temporal trends but smooths out extreme changes.

In addition, a smart energy prediction model is created which combines data pre-processing, prediction of models, and insights for making decisions about energy utilization. It is useful for real-life applications like smart grids, energy planning in industries, and policymaking. Generally, this research emphasizes the significance of choosing suitable models depending on data properties and can be used for short-term as well as long-term energy predictions.

Index Terms—Energy Forecasting, Machine Learning, LSTM, Time Series, Data Analysis

I. INTRODUCTION

Energy usage is an important component in the growth of any economy or society. As more cities expand and the population grows, the demand for energy has also increased rapidly over time. Thus, it is very crucial to monitor and predict energy usage effectively in order to make effective use of resources and reduce costs. Data analysis plays a vital role in this scenario as well.

Though energy consumption prediction is very critical, it can be quite difficult to achieve. There are many variables that

influence energy consumption, such as seasonality, weather conditions, socio-economic activities, festive periods, and geographic variations, among others.

For solving these problems, machine learning algorithms, as well as deep learning methods, have proven their efficacy in the context of energy forecasting tasks. In this regard, machine learning algorithms like Linear Regression, Decision Tree, and Random Forest are useful for analyzing the structure of relations inside a dataset, whereas deep learning models, especially the LSTM architecture, are efficient in modeling temporal dynamics in time-series data. At the same time, the performance of these models can significantly depend on the nature of a dataset and the features it contains.

Apart from that, the majority of the research done currently focuses only on one source of energy, such as electricity consumption, whereas there is no consideration for the multi-source analysis of energy.

These issues are addressed in the current study through a comprehensive analysis of electricity as well as fuel consumption datasets. While electricity dataset ranging from 2013 to 2024 with daily frequency is used for short term prediction, the fuel consumption dataset ranges between 1998 and 2025 with monthly frequency. Further, advanced techniques of exploratory data analysis have been done using methods like seasonal analysis, festive analysis, and state wise consumption analysis to factor real-world influences on energy use.

The contributions of this study can be summarized as:

- Comprehensive analysis of multi-source energy data, including electricity and fuel consumption datasets.
- Real-world considerations like seasonality, festive periods and state wise energy consumption considered.
- Evaluation of machine learning models for predicting energy consumption.
- Implementation of an LSTM model for temporal analysis of energy datasets.
- Formulation of smart energy consumption prediction system.

The main focus of this work is the comparison between various prediction models and the creation of an effective model framework not only able to forecast energy consumption but also to offer valuable conclusions for decision-making processes. The proposed model could be used in various areas

like smart grid operations, energy planning in industries, and sustainable energy optimization.

The rest of this paper is structured in the following way: Literature Review is discussed in Section II, Dataset description is presented in Section III, Data Preprocessing and Exploratory Data Analysis in Section IV, Methodology in Section V, Results and Discussion in Section VI, and Conclusions are offered in Section VII.

II. LITERATURE REVIEW

Energy usage forecast has attracted extensive research efforts because of the significance of energy usage forecasts in energy planning and sustainable development. Many forecasting approaches have been suggested in the literature, including classical approaches, machine learning approaches, and even more sophisticated deep learning techniques.

Initial works were mainly focused on statistical models like ARIMA (autoregressive integrated moving average). Although the models proved themselves powerful for modeling simple trends, they faced difficulties while working with more complex non-linear patterns found in real-life energy datasets.

Since the emergence of machine learning algorithms, Linear Regression has become popular as an approach to solve energy forecasting tasks. Although the Linear Regression algorithm is simple, it has proven itself to be effective in cases where there exist obvious linear correlations between variables. Other types of regression models, Decision Trees, have been considered in order to capture non-linear dependencies and interactions.

In the recent past, many efforts have been made to use deep learning methods to forecast time-series values. RNNs and their variations such as the LSTM neural networks have proved effective in modeling and capturing dependencies in sequential data. Thus, it makes sense that LSTM networks will be effective in energy consumption forecasting due to their ability to capture long-term dependencies in time series data.

Many researchers have proven the applicability of LSTM models to energy consumption prediction. Nevertheless, LSTM networks require substantial data for training and also require the tuning of model hyperparameters. In addition, deep learning algorithms do not necessarily offer better results compared to simpler models such as regression in case data is linear.

The impact of environmental and external factors on energy consumption has lately become a topic of interest. External factors such as climate and social economic activities have a huge impact on energy consumption patterns.

In spite of a vast amount of research in this area, most of the literature focuses on a specific type of energy data, such as the usage of electricity, without paying attention to a multi-data approach. In addition, little effort has been made in terms of cultural and season effects, for example, festivals on energy consumption.

This research paper takes an innovative approach toward studying energy prediction using machine learning and deep learning methods in combination with a complete exploratory

data analysis process. Contrary to conventional methods, this paper considers seasonality effects, festival effects, and even state-wise analysis to reflect real life situations. Moreover, it considers the analysis of electricity consumption data in the short term as well as fuel consumption data in the long term.

The results obtained from this research project show the high dependency of the performance of a model on its dataset, proving once again the importance of choosing the right approach.

A. Research Gap

According to the literature survey, the following research gaps have been found:

- Majority of the research work has been done on a single type of energy source, mainly electricity, but not on multiple sources of energy.
- Very few researchers have studied how real-time events like festivals and seasons affect energy consumption.
- Several methodologies have been used that involve complex models (for instance, deep learning), but they have failed to analyze if simpler models can produce better results for certain datasets.
- There are no combined models that include prediction and its application.

B. Proposed Approach

To fill this void, this research proposes a framework that combines both machine learning as well as deep learning methods for energy prediction purposes. In contrast to current methodologies, the framework uses multiple data sources for the analysis of electricity and fuel consumption.

Moreover, the study involves an advanced stage of exploratory data analysis like season-wise, festival-wise, and state-wise analysis of real-life consumption pattern. Comparative model assessment is also carried out to decide which is the best model according to the data available.

A smart energy prediction system is also proposed in this study to overcome the limitation of converting energy prediction results into practice.

III. DATASET DESCRIPTION

This research relies on two separate data sets to conduct energy consumption analysis; namely, the electricity consumption data set for short-term analysis and the fuel consumption data set for long-term trend analysis.

A. Electricity Dataset

The electricity data set consists of the daily energy consumption values ranging from January 2013 to September 2024. This data set is indexed by time and offers highly detailed records for conducting time series analysis and short-term forecasting. Each record in the electricity data set reflects the total electricity consumed during a certain day. This data set includes the following key variables:

- **Date:** Timestamp indicating the observation date

- **Total Consumption:** Amount of daily electricity consumption.

Because of its daily level of aggregation, this dataset is most effective in recording short-term fluctuations, such as seasonal changes, weekday effect, and rapid increases in consumption. This is the main dataset used for the training of machine learning models and LSTM for prediction purposes.

B. Fuel Dataset

Fuel dataset records the monthly consumption levels of major fuels that include Liquefied Petroleum Gas (LPG), Petrol (MS), and Diesel (HSD). The dataset covers a long period of time between April 1998 to March 2025.

Dataset characteristics are as follows:

- **Products:** Name of the fuel (LPG, Petrol, Diesel)
- **Year:** Year of the observation
- **Month:** Month of observation
- **Month Name:** Name of the month
- **Consumption in Metric Tonne:** Consumption levels of the fuel

This dataset can be mainly used for trend analysis and comparative studies of different fuels. Unlike electricity dataset, it contains monthly values.

C. Dataset Integration Challenges

One of the main problems that arise in relation to these datasets is the disparity in their levels of granularity. While the electricity dataset is collected on a daily basis, the fuel dataset is collected on a monthly basis. This problem hinders the integration of both datasets for the purpose of modeling.

Furthermore, these two datasets are not homogeneous with regard to their structure and size. The amount of electricity consumed is measured on a daily basis and is thus expressed in terms of consumption per day. In contrast, the amount of fuel consumed is measured in metric tons.

Because of these problems, the datasets are modeled individually but used in joint interpretation in the exploratory stage.

D. Data Sources

The electricity consumption data that will be used in this study is freely available on Kaggle, and it is collected from POSOCO, Government of India. The dataset comprises daily electricity demand data for different states and UTs.

The fuel consumption data was collected from the respective governmental websites and is available in PPAC, Ministry of Petroleum and Natural Gas, Government of India. The data comprises monthly consumption of various petroleum products, including LPG, Petrol (MS), and Diesel (HSD).

These datasets are available at the following URLs:

- Electricity Dataset:
<https://www.kaggle.com/datasets/aryankhurana1701/state-wise-electricity-consumption-in-india>
- Fuel Dataset (Government Source):
<https://ppac.gov.in/index.php/consumption/products-wise>

IV. DATA PREPROCESSING

Data preprocessing plays an important role in the data analysis process since it is responsible for providing high-quality, consistent, and usable data. Unprocessed data usually have issues such as missing values, inconsistency in formatting, and noise. In this project, several types of data preprocessing will be performed on electricity and fuel data sets.

A. Handling Missing Values

Missing data are detected and taken care of to keep the data clean. If the feature is numerical, then missing data will be imputed using techniques like mean imputation, or they can simply be discarded.

B. Date-Time Conversion

Columns containing dates are formatted to a uniform datetime format, enabling time series analysis by making it easier to sort, index, and extract information such as month and year.

C. Indexing for Time-Series Analysis

The datetime column serves as the index of the data frame. Such an operation becomes essential for carrying out time series-related operations like rolling mean, trends, and time-based splits for model training and validation.

D. Feature Engineering

More features are derived from the datetime index to further enhance the performance of the model. They are:

- **Month:** Accounts for seasonality in energy consumption
- **Year:** Enables analysis of trends over an extended period
- **Day/Week Patterns (if applicable):** Accounts for short-term fluctuations

Feature engineering enhances the model's ability to learn meaningful patterns from the data.

E. Rolling Average Calculation

The rolling average is calculated to remove the short term volatility and emphasize the long term trend. It is especially beneficial for depicting consumption trends from the data set.

F. Categorical Encoding

Categorical variables like festivals' indicators can be converted using methods like one-hot encoding. Thus, the machine learning algorithms can take into account these categorical variables when making predictions.

G. Data Normalization

As for neural network approaches like LSTM, data normalization is carried out using various scaling methods (such as min-max scaling). Data normalization helps to standardize input values so that they lie on the same scale.

H. Train-Test Split

This data set has been split into training and testing data sets for evaluation purposes. Because this dataset depends on time, a time-based split is preferred rather than random sampling.

I. Outlier Detection and Handling

The energy consumption figures may have some unexpected values that result from environmental influences, such as unusual weather patterns. Such anomalies will be detected through visual methods as well as mathematical methods. They will then either be kept as part of the training set or altered depending on their effect.

J. Data Alignment and Aggregation

As the data on electricity is recorded on a daily basis and that of fuel is recorded on a monthly basis, the data cannot be merged directly. In such cases, aggregation methods can be used to make comparisons between different data sets. The datasets will be modeled separately, whereas alignment will be done during exploratory data analysis.

K. Feature Selection

The features that are relevant to the target feature are chosen according to their relevance to the target feature. The irrelevant features are eliminated since they will add to the complexity of the model without improving its performance.

L. Data Consistency Check

The consistency test is carried out to confirm whether the data is free from any duplications, wrong entries, or inconsistency among the variables. The integrity of the data is crucial in making accurate projections.

V. METHODOLOGY

The overall methodology applied in the current project is depicted. This methodology is representative of the entire process, starting from data preprocessing to system development.

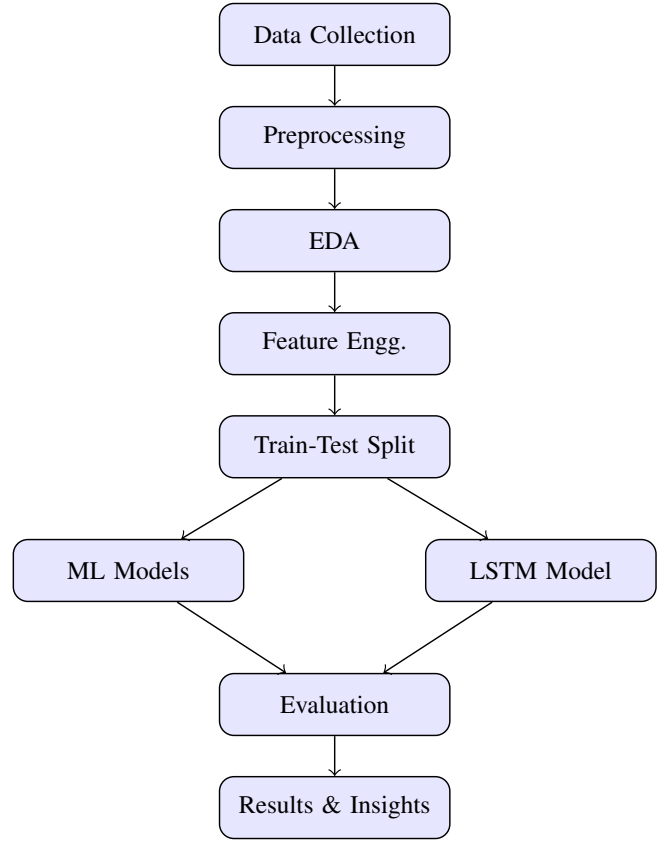


Fig. 1. Methodology Flowchart of the Proposed Approach

In the current research project, empirical research methods that are driven by data will be adopted. The methodology used in this research is depicted by the sequential pipeline of various stages.

A. Data Preparation

Dataset: The dataset comprises past consumption rates of electricity of different states along with other attributes, including fuel consumption (LPG, petrol, and diesel). Data preprocessing is done so as to ensure that the data set is authentic and reliable. Missing values in numeric data are replaced with the mean value, whereas one-hot encoding is done for categorical data (seasons and festivals). Moreover, time variables like days, months, and years are created.

B. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is conducted to understand the structure and behavior of electricity consumption. This includes:

- Trend analysis to identify long-term growth patterns
- Monthly and seasonal analysis to capture periodic variations
- Festival-based analysis to detect short-term demand spikes
- State-wise analysis to understand regional consumption differences

- Fuel-wise analysis to examine sector-specific energy usage
- Correlation analysis to identify relationships between variables

These analyses provide key insights that guide feature selection and model development.

C. Train-Test Split

As the data set is of time series type, hence splitting is done sequentially rather than randomly. Around 80% of the data is utilized for training purposes, whereas 20% is kept aside for testing.

It makes sure that the model is validated using future data, thereby making the evaluation process more authentic.

D. Model Development

Multiple machine learning models are implemented to capture different types of relationships within the data:

- Linear Regression: Serves as a baseline model to capture linear relationships.
- Decision Tree Regressor: Captures non-linear and hierarchical patterns.
- Random Forest Regressor: An ensemble model that improves accuracy and reduces overfitting.

Each model is trained on the training dataset and used to generate predictions on the test dataset.

E. Deep Learning Approach

In order to enhance the forecast accuracy even more, an LSTM model is developed. The use of LSTM models for time series data is very appropriate, as they have the ability to identify long-term dependencies within sequences.

This means that temporal dependencies are taken into account that may otherwise be ignored.

F. Model Evaluation

The performance of all models is evaluated using multiple metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R-squared (R^2)
- Mean Absolute Percentage Error (MAPE)

These metrics provide a comprehensive evaluation of prediction accuracy and enable objective comparison between models.

G. System Design

According to the findings gained from the analysis of data and model performance, there emerges a need for developing a system for predicting power consumption. This system will help to make predictions, minimize inefficiencies, and plan for energy requirements.

The following sections present the detailed analysis and results based on the methodology described above.

VI. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is performed to understand underlying patterns, trends, and relationships in the data. It helps in identifying seasonal effects, anomalies, and important features that influence energy consumption.

A. Electricity Trend Analysis

It is important to know the trends in the way the amount of electricity used changes with time since this information can help in forecasting future trends. This trend is illustrated by plotting the electricity consumption against time with both the actual value and moving average.

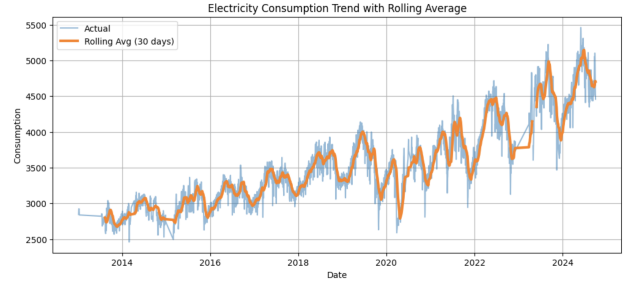


Fig. 2. Electricity Consumption Trend with Rolling Average

Initially, the trend of electricity consumption is very difficult to determine since the actual consumption figures are volatile, experiencing short-term increases and declines. It is normal in real-life statistics that each day's energy consumption is affected by different variables such as climatic conditions, social events, and human activities. Therefore, though the actual consumption figures give insight into the real behavior, they make it hard to recognize the actual trend.

In order to mitigate this problem, a 30-day rolling average is used. By calculating the 30-day average over different periods of time, it becomes possible to eliminate short-term irregularities that may obscure the actual trend. As such, the rolling average gives us an indicator of how the actual energy consumption evolves over time.

The resulting graph shows that there is a rising trend in electricity consumption over time. This increasing trend can be explained by several factors, including fast urbanization, growing populations, greater use of electrical appliances, and rapid economic development, among others. This growth is progressive rather than explosive. In addition to the upward trend, the data also exhibits recurring cyclical patterns. These cycles indicate seasonal variations in electricity usage, where certain periods consistently show higher or lower consumption. For instance, peaks in the rolling average curve may correspond to summer months when cooling demand is high, while dips may align with relatively moderate seasons.

An additional interesting pattern is sharp deviations of the actual consumption values from the rolling average

value. This kind of anomaly shows the influence of some temporary changes or external factors that can cause short-time deviations in electricity consumption. Distinguishing between short-term anomalies and real data is very important, and this example can serve as an illustration of this issue.

The above discussion shows that electricity consumption has at least three main parts: the long-run trend, the seasonal component, and short-run changes. In order to forecast electricity consumption, one should first understand the structure of historical data in order to choose proper models. Although machine learning algorithms can help to discover patterns, it is more convenient to use time-series models like LSTM.

Trend analysis is only a preliminary step before developing a proper forecasting algorithm.

B. Monthly Analysis

Even though the total trend of electricity consumption gives a general idea of rising demand, it does not clearly define the reasons for the observed variations in the data. For this purpose, the monthly variation analysis is carried out by adding up the values of consumption per month from different years.

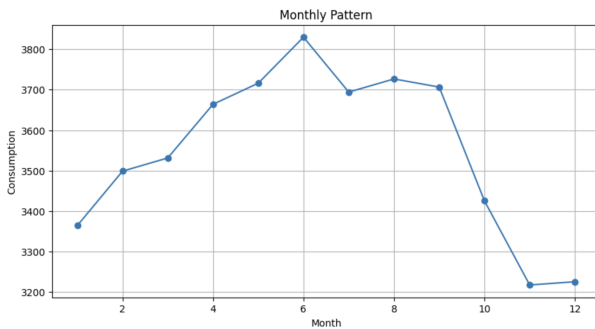


Fig. 3. Month-wise Electricity Consumption

From the findings presented in the analysis above, there is a definite seasonal trend in electricity consumption. From the graph, it is evident that the consumption of electricity starts to increase steadily in the months of January to March. The increase in consumption becomes steeper in April, May, and June months. The maximum electricity consumption occurs in the months of May and June.

It is from this finding that the previous analysis on the trend chart and rolling average became clearer. There are repeated peaks in the trend chart that coincide with the summer season. Due to the higher temperatures, there is a significant increase in the use of appliances such as fans, air conditioning units, and refrigerators. Therefore, the monthly analysis presents detailed information concerning the cycles observed in the trend analysis.

Once the peak season comes to an end, there is a small reduction in consumption in the July, August, and September months. During this period, the weather

becomes relatively cool; however, due to humidity, the consumption is still considerable. That is the reason for the small dip in the rolling average trend in these months. It is interesting to note that the fall becomes more apparent towards the latter part of the year, namely between the months of October and December. The weather is much milder in this period, and there is no need for air conditioning systems to be used, thereby reducing the rate of consumption.

One important aspect to note here is that the same pattern repeats every year. This recurring cycle shows that the use of electricity is highly dependent on seasonal variables rather than any arbitrary factors. Moreover, the cyclic nature of the rolling average line reflects this aspect as well.

Overall, it can be concluded that there are some significant seasonal dependences in electricity consumption revealed during the monthly analysis. The importance of this finding lies in its use within the trend analysis process to explain any changes observed in consumption and also for forecasting purposes. It should be emphasized that considering attributes such as month and season in the model under analysis is very important due to their significance in forecasting energy consumption.

This becomes especially important in developing the prediction models with respect to time series, including LSTM methods.

C. Seasonal Analysis

Though the monthly calculation offers an elaborate description of electricity usage in each month, it might be too complicated to recognize patterns at times. To make things easier, the values are categorized by seasons, thus making it easier to understand consumption trends.

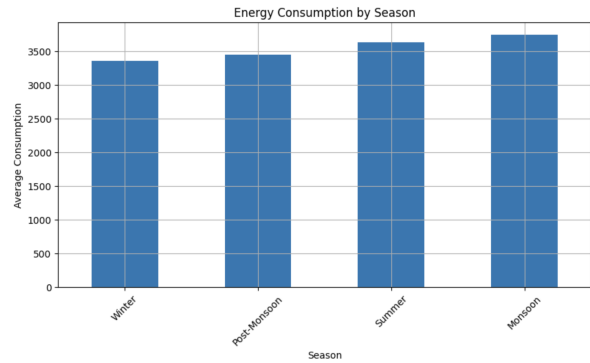


Fig. 4. Seasonal Energy Consumption

From the figure below, it is evident that there are differences in the consumption of electricity based on different seasons. From among all the seasons, the period when there are relatively high average consumption rates is during the monsoon and summer seasons; the lowest consumption rates, however, are noted in winter. Post-

monsoon has moderate consumption of electricity and acts as a transitional period.

Based on these findings, it becomes clear that the observations made during the monthly analysis are supported by what was found using the grouped data approach. Earlier, it was noted that consumption is highest during the months of summer (May and June), and this can be noted through the higher average consumption rate during the summer season. Similarly, the trend witnessed at the end of the month where electricity use fell can be attributed to the lower use of energy in the winter season. Grouping by seasons also helps in understanding why there were cycles noted in the electricity trend and rolling average graph. The high points in the graph can be attributed to the high-demand seasons like summer and monsoon; the low points, on the other hand, can be attributed to low-demand periods like winter.

An interesting thing to note from the graph is that there is still considerable usage during the monsoon months despite the fact that the temperature is much cooler than during the peak summer months. This is primarily due to the fact that people spend more time indoors, use more humidifiers, and continue using electrical appliances throughout the year.

In general, it is evident from the seasonality analysis that electricity usage is not haphazard but rather exhibits a cyclic pattern based on climatic conditions. This knowledge is important for prediction because it emphasizes the necessity of accounting for seasonality in predicting models. An example of such a model is the LSTM model, which is capable of learning the recurrent seasonal relationships within data.

Therefore, seasonality analysis provides an intermediary step between detailed monthly analysis and long-term trends.

D. Festival Analysis

Apart from these factors, energy consumption is also affected by various social-cultural activities. The festivals are one of the most significant factors that affect the energy demand for a short period because of the increase in human activity during festive seasons.

To include this factor into consideration, a feature based on festivals is included in the data. However, rather than considering exact days when these festivals take place, as these days are different each year and are therefore hard to track, a month-wise consideration is taken up. For example, March, September, Navratri, and October/November are considered festivals as they include Holi, Ganesh Chaturthi, and Diwali, respectively.

With this feature, the effects of major festivals are included without any problem of inconsistency as well as without including any trivial festivals that may create noise.

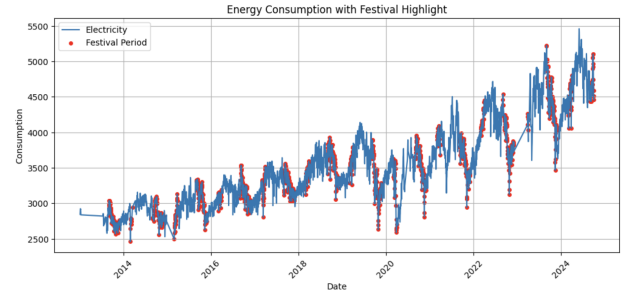


Fig. 5. Energy Consumption with Festival Highlights

It can be seen from Figure 5 that the red dots signify the festival periods detected. It can be seen that some of these dots match certain upward spikes or deviations in electricity consumption relative to other days around it. It shows that festival periods have temporary effects on increased energy demands.

This can be easily justified by looking at real-life conditions, wherein during the period of Diwali there will be considerable lighting, whereas Navratri and Ganesh Chaturthi will be occasions where there will be large gatherings and increased activities. In addition, even during Holi there will be considerable social mobility.

The variations due to festival periods help understand the anomalies that occur after considering the seasonal variation in electricity consumption. While the former helps in understanding the periodic changes in consumption, the latter introduces temporary anomalies that can easily be noticed by observing the original graph but get leveled out in the case of a rolling average.

Also worth noting is the fact that these festival spikes are localized in time and do not have a substantial effect on the general trend of electricity consumption growth. Nevertheless, they are extremely significant when it comes to making forecasts and managing demand, since they may create some extra burden on the system temporarily.

In general, it has been found that electricity consumption depends not only on environmental factors like seasonality but also on human behavior and other aspects of culture. Taking into account information about festivals makes the forecasting process much more realistic.

E. State-wise Energy Consumption Analysis

In order to have an enhanced knowledge of the regional patterns of electricity demand, a geographical study is undertaken based on the calculation of the mean electricity consumption per state throughout the entire duration of time. This differs from the temporal method (monthly/seasonal) since it emphasizes geographical variation more.

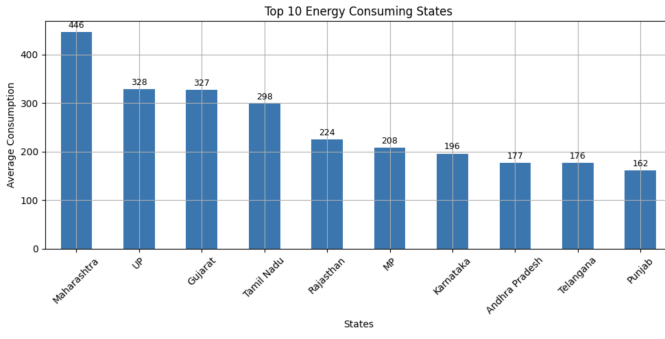


Fig. 6. Top 10 Energy Consuming States (Average Consumption)

From the graph, it shows the top 10 states ranked according to their mean power consumption. It is evident that there is a distinct difference between these rankings, suggesting that there is unequal energy consumption among different states.

Firstly, Maharashtra ranks highest in terms of energy consumption due to its higher consumption than other states. This is primarily because of its dense population, industries, and presence of major cities like Mumbai and Pune, which require constant power consumption.

Following Maharashtra, we have Uttar Pradesh and Gujarat. The main reason for the higher power consumption in Uttar Pradesh is its huge population size. On the other hand, the main contributor to power consumption in Gujarat is its highly developed industrial and manufacturing sector.

There is high consumption of electricity in Tamil Nadu and Rajasthan. Tamil Nadu has been described to be an industrial state. In Rajasthan, however, this has been attributed to climatic conditions where there are high temperatures causing cooling requirements. The consumption of electricity in Madhya Pradesh and Karnataka is at medium level, implying that there is equal consumption in residential areas and industries in those states. Consumption of electricity in Andhra Pradesh, Telangana, and Punjab is relatively low despite being among the ten best consuming states.

From this analysis, one major finding is that the variance between the highest and the lowest consumption in the ten best consuming states is very high. There is, therefore, high variation in electricity consumption levels depending on several determinants such as population and climatic factors.

Such observations have a number of practical applications with respect to energy management. For example, states with high electricity consumption levels might need upgraded electrical infrastructure, improved demand management, and load balancing solutions. At the same time, those with low levels of consumption could represent potential areas for the future development of infrastructure and its improvement.

Moreover, such discrepancies across different regions

show that the use of state variations is a vital component in the forecasting process. While a unified forecasting model might not reflect these differences, region-specific information can significantly increase its accuracy and effectiveness.

Finally, this analysis of the electricity consumption at the level of individual states supplements the previously conducted analyses that focused on periods of time. All of them are crucial in developing forecasts for electricity consumption and designing a corresponding management system.

This indicates the importance of localized energy forecasting models.

F. Fuel-wise Consumption Analysis

Although there is important information that can be gleaned from electricity usage, this cannot alone provide a comprehensive view of energy usage. This is because a substantial amount of energy usage comes from other sources of energy, including LPG, petrol, and diesel. This means that by considering fuel usage, we get a more comprehensive picture of energy usage.

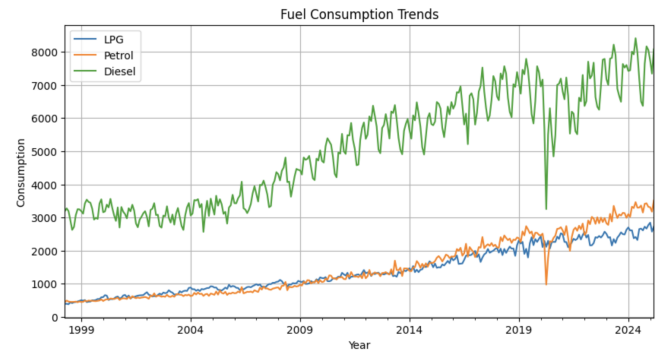


Fig. 7. Fuel Consumption Trends Over Time

The graph depicts the consumption patterns of LPG, petrol, and diesel within the time frame under observation. The positive trend of increased consumption is apparent for all three energy sources, suggesting that the total energy consumption is continuously rising.

The consumption of diesel is relatively higher compared to the other two energy sources through the entire time frame. It is mostly because of the high demand from transportation, agricultural, and industrial activities. Moreover, the positive growth trajectory indicates a continuous increase in the economy and construction activities. A drop in consumption is noted around 2020, which is because of low industrial production and transportation due to the outbreak of the novel coronavirus.

The consumption of petrol is also growing positively with a consistent increase over the years. As for the comparison with diesel, the consumption of petrol is comparatively less stable with some volatility based on mobility and economic factors. Also, a similar dip in

consumption was recorded around 2020 before recovering again.

The growth in consumption of LPG has been comparatively smooth and steady. It has largely been attributed to the wide use of this fuel in households for purposes such as cooking. The government policies advocating for the use of environmentally friendly fuels have also aided the stability in the consumption trend. In comparison to other types of fuels, LPG experiences lower volatility due to its stability.

One significant aspect that emerges from the analysis above is the difference in the volatility of various fuels. The higher volatility of diesel is attributed to the fact that its consumption relies heavily on industrial/commercial activities, unlike LPG, which is constantly consumed at households. Petrol has volatility that falls in-between those of diesel and LPG.

In conclusion, the fuel-wise analysis indicates the different nature of energy consumption by varying sectors. All the fuels experience growth over time, but their variations vary depending on their usage characteristics. Therefore, it is essential to include insights from various fuel consumptions in any analysis regarding energy consumption patterns.

Moreover, integrating insights from the fuel-based approach can enhance prediction through improved accuracy and hence proper design of an energy management system.

G. Correlation Analysis

For more insights into the interconnection between various energy elements, correlation analysis is carried out between the total electricity usage and fuel usage indicators like LPG, petrol, and diesel.

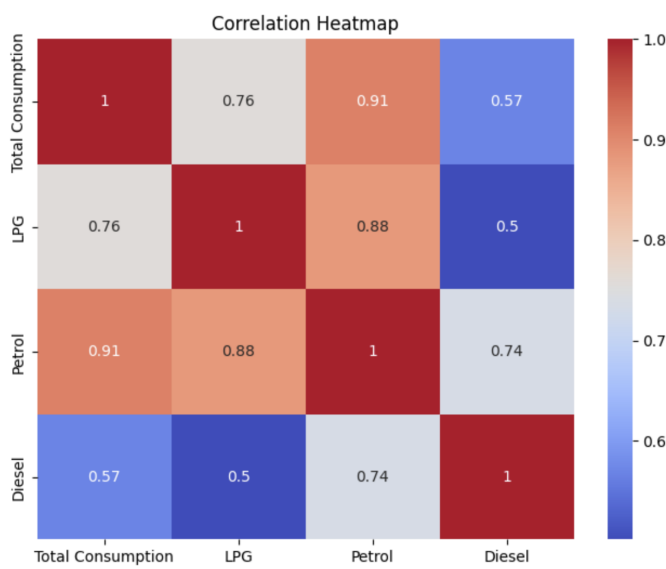


Fig. 8. Correlation Heatmap between Energy Variables

In the heatmap above, we can see the extent of the linear correlations between the indicators. We see there is a high correlation between the total consumption and petrol consumption (0.91). The fact implies that petrol consumption follows the same trends as the overall consumption pattern. Therefore, transportation plays an important role in total energy consumption.

Similarly, LPG demonstrates a moderately high correlation (0.76) with total consumption due to the fact that this fuel is commonly used in residential sectors. The regular growth of LPG is constantly adding up to overall energy consumption.

At the same time, the correlation of diesel with total consumption is noticeably lower (0.57). Despite the fact that diesel consumption is quite high in general terms, this type of energy consumption is still characterized by some variability that is determined by industry and agriculture. Furthermore, a strong correlation (0.88) between petrol and LPG demonstrates the possibility of the simultaneous rise of both of these types of fuel. It may be explained by the parallel development of economic sectors that require them.

On the whole, the information about correlation helps to understand the relationship between different types of energy consumption. It is especially important in the context of selecting features used in predicting energy consumption in the future.

This correlation analysis also helps in feature selection for machine learning models.

VII. DATA PREPARATION AND PREPROCESSING

After gaining insights from exploratory data analysis, the next step involves preparing the data for model building.

1) *Train-Test Split*: Since the data under analysis is time series data, it is important that the temporal sequence be retained throughout the process of splitting. Random sampling cannot be used in this case; instead, sequential splitting would be employed.

The data would be divided between the training and testing datasets, such that about 80% of the data would be used for training purposes, while the other 20% would be used for testing purposes.

2) *Feature Engineering and Preprocessing*: The model uses both temporal and context features obtained from preprocessing. The features include:

- Time-related features like day, month, year, and day of the week
- Lag-related and rolling statistics-based features related to historical consumptions
- Exogenous features like the consumption of fuels (LPG, petrol, diesel)
- Holiday features depicting demand variation on account of festivities

The one-hot encoding method is used to convert categorical features like season and holiday to be compatible

with the machine learning algorithm. Mean imputation is used for numerical features.

Preprocessing guarantees that data will be ready for modeling.

A. Machine Learning Models

To evaluate different modeling approaches, multiple machine learning algorithms are implemented to predict electricity consumption based on the prepared dataset.

1) *Model Selection*: The following models are considered:

- Linear Regression: A baseline model used to capture linear relationships between features and electricity consumption.
- Decision Tree Regressor: A non-linear model capable of capturing complex and hierarchical patterns in the data.
- Random Forest Regressor: An ensemble learning method that combines multiple decision trees to improve accuracy and reduces overfitting.

These models are selected to compare linear and non-linear learning capabilities and to identify the most suitable approach for the given dataset.

B. Model Training

Training of all models is done using the training data set, wherein they learn how the input features are correlated to electricity usage. Trained models are further used for making predictions on the testing data set.

1) *Evaluation Metrics*: To assess model performance, the following evaluation metrics are used:

- Mean Absolute Error (MAE): Measures the average magnitude of prediction errors.
- Root Mean Squared Error (RMSE): Penalizes larger errors more heavily and highlights significant deviations.
- R-squared (R^2): Indicates how well the model explains the variance in the data.
- Mean Absolute Percentage Error (MAPE): Provides a relative measure of prediction accuracy in percentage terms.

These metrics provide a comprehensive evaluation of model performance and enable objective comparison between different models. The results obtained from these metrics are discussed in the following section.

C. Model Evaluation and Results

The performance of the machine learning models is evaluated using multiple metrics, including MAE, RMSE, R^2 , and MAPE. The results are summarized in Table I.

TABLE I
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Model	MAE	RMSE	R^2	MAPE
Linear Regression	26.22	29.60	0.994	0.0058
Decision Tree	288.29	348.39	0.241	0.063
Random Forest	278.75	341.24	0.272	0.060

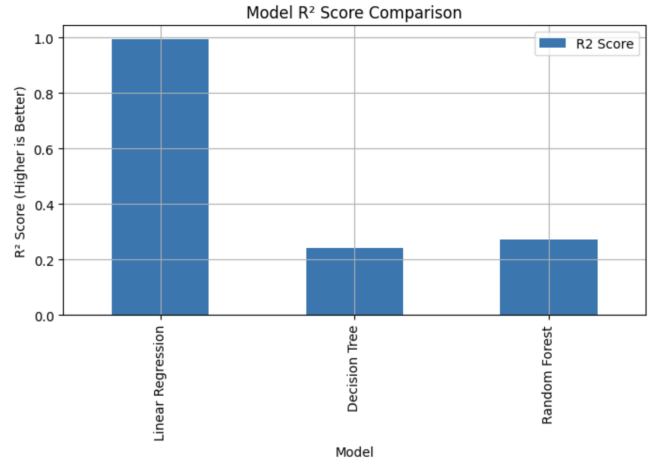


Fig. 9. Comparison of R^2 Scores Across Machine Learning Models

The figure given above depicts a comparison among R^2 measures for various machine learning models. As seen from the results, the linear regression method performs better with the highest value of the measure, which suggests that this model outperforms both Decision Tree and Random Forest approaches.

Based on the results provided above, one may conclude that linear regression demonstrates better performance compared to other machine learning algorithms with respect to all evaluation metrics. Specifically, it obtains the minimum values of both MAE and RMSE measures together with a highly significant value of the R^2 measure, thus ensuring a good fit.

In addition, the values of error measures in the case of Decision Tree and Random Forest approaches are higher compared to linear regression, and the R^2 value is lower, meaning that the first two models perform poorly in terms of generalization ability.

The choice of linear regression may suggest that the dependence between variables is predominantly linear; this corresponds to patterns revealed in the course of exploratory data analysis.

As a result, the linear regression algorithm is used to solve the task described above.

However, despite such strong results, the linear regression model does not consider temporal dependences inherent in the given type of dataset.

In order to overcome this drawback, an LSTM model will be utilized in the subsequent section, since it is well suited for detecting temporal dynamics and long-range dependencies.

Poor performance of decision trees suggests overfitting and lack of generalization ability for linear data.

D. Deep Learning Approach: LSTM

Even though linear regression exhibits excellent performance, it lacks the ability to represent temporal dependency existing in time series data. The electricity usage

is indeed sequential data, where the past affects future values.

This drawback is addressed through an LSTM model. LSTM is a special kind of RNN used for modeling long term dependencies in sequential data. It is particularly suited for our problem due to its capacity to remember past values.

1) *Data Preparation for LSTM:* Preprocessing: Before training the LSTM model, data preprocessing is performed to structure the information as a time series. This is achieved by generating sequences based on past values of electricity consumption.

Furthermore, the process of feature scaling is carried out to scale the features of the dataset.

2) *Model Architecture:* The LSTM model consists of sequential layers that process time-series data and learn temporal patterns. The architecture typically includes:

- LSTM layers to capture sequential dependencies
- Dense layers to generate final predictions

The model is trained using the training dataset and optimized to minimize prediction error.

3) *Model Evaluation:* The performance of the LSTM model is measured using the same performance metrics used by the machine learning models, namely MAE and RMSE. This creates a level playing field between the two methodologies.

TABLE II
LSTM MODEL PERFORMANCE

Model	MAE	RMSE	R ²
LSTM	105.40	129.36	0.896

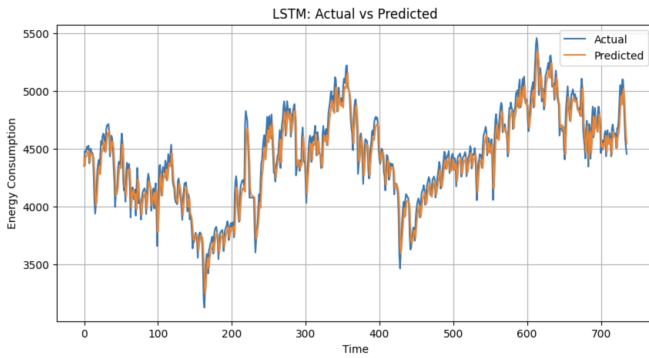


Fig. 10. Actual vs Predicted Electricity Consumption using LSTM

The following diagram demonstrates the comparison between the actual and forecasted power usage through the LSTM model. It can be seen that the predicted values follow the actual ones, meaning that the model is capable of capturing the temporal patterns of the dataset.

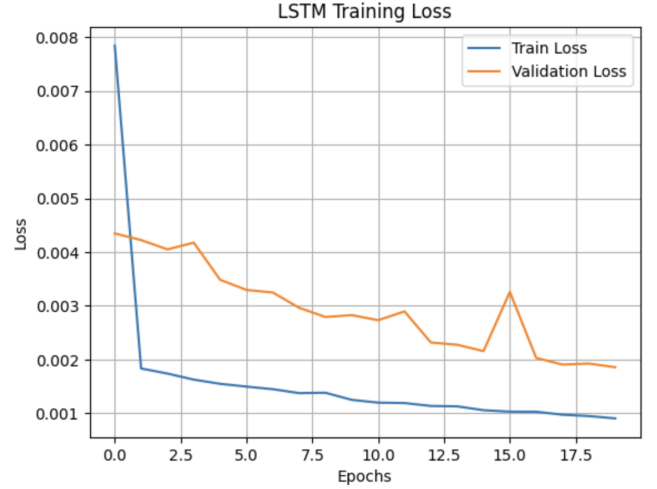


Fig. 11. Training and Validation Loss of LSTM Model

As can be seen from the loss curve, there is a drop in the training and validation loss as the number of epochs progresses. Consistent decreases in the value of loss suggest that learning was successfully implemented, whereas low discrepancy between training and validation loss implies that there is no serious overfitting problem. Comparing the LSTM model to other models, one can see that it demonstrates better performance by yielding error values several times smaller than those of decision trees and random forests along with a greater R² coefficient. Nevertheless, comparing the LSTM model to linear regression, one can conclude that it is inferior to the latter since it produces somewhat larger errors along with a lower R². As a result, linear regression is the best model of all tested ones.

From the superior results yielded by linear regression, one may conclude that the dataset contains mostly linear relationships.

This provides a critical insight: more complex models do not necessarily translate to better results. Even though LSTM works well for modeling sequential data, simpler models like linear regression might yield superior results to deep learning algorithms when the underlying patterns in the data are mostly linear.

Thus, models should be selected based on their suitability with the type of data and not on how complex they are. LSTM tends to average out predictions, leading to slightly less accurate predictions compared to linear regression.

VIII. PROPOSED SYSTEM

The design and implementation of this system will involve the use of machine learning and deep learning in order to analyze data and predict future energy consumption.

A. System Overview

Historical data regarding energy consumption is fed into the system, which analyzes and extracts key features from

this data in order to make predictions about the future.

B. System Architecture

The system consists of the following key components:

- **Data Collection:** Electricity and fuel consumption datasets are collected from reliable sources. The electricity dataset contains daily data, while the fuel dataset contains monthly data.
- **Data Preprocessing:** Missing values are handled, date formats are standardized, and relevant features such as month, year, and season are extracted.
- **Exploratory Data Analysis:** Various analyses such as trend, monthly, seasonal, and state-wise patterns are performed to understand the data.
- **Feature Engineering:** Additional features such as rolling averages and time-based attributes are created to enhance model performance.
- **Model Training:** Machine learning models including Linear Regression, Decision Tree, and Random Forest are trained on the processed data.
- **Deep Learning Model:** An LSTM model is implemented to capture temporal dependencies in the time-series data.
- **Model Evaluation:** Models are evaluated using performance metrics such as MAE, RMSE, and R2 Score.
- **Prediction and Insights:** The trained models are used to generate predictions and provide insights into future energy consumption trends.

C. System Workflow

The workflow of the system is illustrated as a sequence of steps:

- 1) Input raw energy consumption data
- 2) Perform data cleaning and preprocessing
- 3) Conduct exploratory data analysis
- 4) Generate features for modeling
- 5) Train machine learning and deep learning models
- 6) Evaluate model performance
- 7) Generate predictions and insights

This structured workflow ensures that the system can efficiently process large datasets and produce accurate and meaningful predictions.

D. Applications of the System

The proposed system can be used in multiple real-world scenarios:

- Energy demand forecasting for power distribution companies
- Resource planning and load management
- Policy making for sustainable energy usage
- Identifying peak demand periods and optimizing energy supply

Overall, the system provides a comprehensive framework for analyzing and predicting energy consumption using

data-driven techniques. This system bridges the gap between data analysis and real-world deployment.

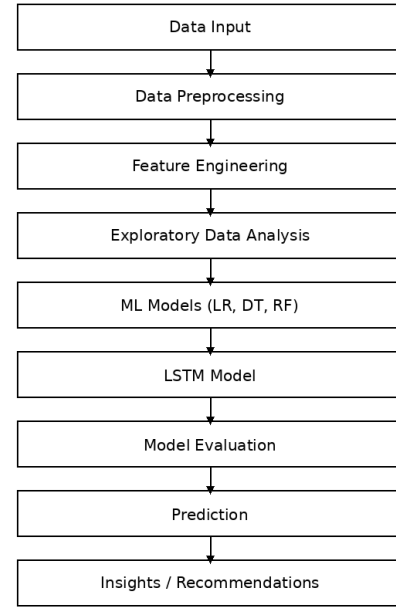


Fig. 12. Proposed System Workflow

IX. RECOMMENDATIONS

From the findings of the analysis and forecast, some recommendations may be suggested to help with energy management and optimal use of energy resources.

A. Demand Management During Peak Periods

The results of the analysis indicate that there is a considerable increase in the use of electrical power at particular times. The peak times of electrical energy consumption were recorded to be in June, whereas the minimum consumption was in November. This is because of climate variations. Higher temperatures cause people to turn on their cooling devices, like air conditioners and fans.

Demand-side management policies should be considered by energy companies in order to control energy consumption peaks. Users should be advised to avoid unnecessary energy consumption during peak hours.

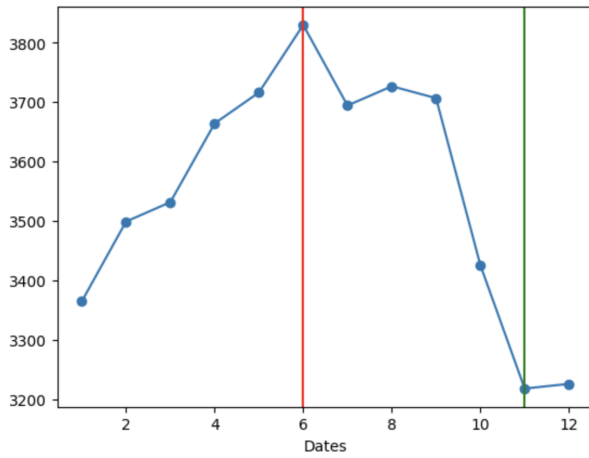


Fig. 13. Peak vs Low Energy Consumption (Monthly Comparison)

B. Peak and Off-Peak Optimization

In addition, there are noticeable distinctions between peak and off-peak consumption amounts. The amount of peak consumption is found to be 3829.59, whereas the amount of lowest consumption is 3217.62, thus having a distinction of 611.97 (around 19 percent).

Such a considerable distinction emphasizes the significance of load management and energy distribution. Electricity suppliers may apply time-of-use policies and demand response to ease the burden on the system during the peak time.

C. Seasonal Energy Planning

The effect of seasonality on energy usage shows that it greatly depends on the weather. The use of more energy during summer months calls for more energy generation and distribution. It is advisable for energy suppliers to keep more stock during high seasons and make use of low seasons for maintenance.

D. Region-specific Resource Allocation

Regional-level assessment shows that there exists a considerable variance in terms of energy consumption across various areas. High density of population, industrialization, and urbanization in a state contribute towards higher consumption of energy.

It would be advisable for the policy makers to employ regional-level measures for the distribution of energy sources.

E. Smart Energy Usage Recommendations

Based on the observed consumption patterns, the following smart energy practices are recommended:

- **Optimized Cooling Usage:** Consumers should limit excessive air conditioner usage and maintain optimal temperature settings between 24 and 26°C to reduce energy consumption during peak summer months.

- **Load Shifting:** High-energy-consuming activities should be shifted to off-peak periods to balance demand and reduce grid stress.
- **Energy-Efficient Appliances:** The use of energy-efficient appliances, such as 5-star rated devices, should be encouraged to minimize overall consumption.
- **Awareness Initiatives:** Public awareness campaigns should be conducted to promote responsible energy usage, especially during high-demand periods.
- **Smart Industrial Scheduling:** Industrial operations should adopt intelligent scheduling techniques to distribute energy usage more evenly across time.

F. Integration of Predictive Systems

The application of machine learning techniques like linear regression and LSTM models proves the usefulness of predictive analytics in predicting energy needs. Energy providers should be encouraged to adopt these technologies as part of their system to monitor energy needs in real time.

G. Sustainable Energy Practices

Considering that there is an upward trend in the use of energy, it is crucial to embrace sustainability through the use of renewable energy resources, energy-saving methods, and effective infrastructure construction.

In summary, these recommendations strive to increase energy efficiency, minimize problems related to energy use, and ensure sustainable energy management based on data analysis.

X. CONCLUSION

In this research work, an extensive analysis and prediction of energy consumption is carried out using both machine learning and deep learning techniques. Using the electricity and fuel consumption dataset, the project aims to analyze the trends both for short periods of time and in the long run.

Exploratory data analysis helped unearth some key findings. Firstly, there was a positive trend witnessed in the electricity consumption dataset, pointing towards rising consumption over time. In the seasonal and monthly data analysis, cyclical patterns were seen, with high energy consumption being observed during the summer season. Moreover, in the festival-based data analysis, short peaks in energy consumption were witnessed owing to increased activities. Lastly, state-wise data analysis helped uncover differences between various states.

From the modeling point of view, several machine learning algorithms were considered, namely linear regression, decision trees, and random forests. Among them, linear regression yielded the highest results because of the linear nature of the dataset used. An alternative model that was also developed is LSTM, as an attempt to detect any temporal dependencies in the data. Even though it

succeeded in detecting sequence patterns in the data, LSTM gave poorer results than linear regression.

In summary, this project showed how data-driven methods could be applied in the sphere of energy forecasting and planning. It demonstrated how data analysis could help predict energy consumption and plan resource allocations accordingly.

This work proved that it was essential to choose an appropriate model depending on the properties of data. Also, the presented project showcased the value of using statistical data analysis along with machine learning algorithms.

As a result, the information gained from this research could be useful in future projects regarding energy consumption forecasting and planning.

XI. SYSTEM DESIGN DIRECTION

The system design can further evolve into an implementable system that effectively manages energy consumption with real-time data processing capabilities, intelligent prediction methods, and intuitive user interfaces. The proposed evolution of the design is to turn the theoretical analysis into a practical system.

A. System Architecture

The future system can be structured into multiple layers:

- **Data Layer:** This layer collects data from multiple sources, including smart meters, IoT devices, weather APIs, and historical energy datasets.
- **Processing Layer:** Data preprocessing, cleaning, and feature engineering are performed in this layer to ensure data consistency and quality.
- **Model Layer:** Machine learning and deep learning models, such as linear regression and LSTM, are used to generate energy consumption forecasts.
- **Application Layer:** This layer presents predictions, insights, and recommendations through dashboards and visualization tools.

B. Workflow of the System

The system operates through the following sequence of steps:

- 1) Data is collected from real-time and historical sources
- 2) Preprocessing and feature extraction are applied
- 3) Forecasting models generate predictions
- 4) Results are evaluated and stored
- 5) Insights and recommendations are presented to users

C. User Interaction

The system can be designed for multiple types of users:

- **Energy Providers:** For load forecasting and grid management
- **Industries:** For optimizing energy usage and reducing costs
- **Government Authorities:** For policy planning and resource allocation

D. Deployment Strategy

The system can be implemented either as a web-based or cloud-based application. The system can have a user interface that will help visualize the consumption pattern, identify peak demands, and provide suggestions for improvement.

E. Scalability and Extension

The architecture can be extended to accommodate various regions and different types of energy as well. Future possible extensions may involve combining the architecture with renewable energy sources, smart grids, and automation systems for energy management.

In conclusion, the future architecture design direction lies in making a robust and intelligent architecture that enables efficient energy management and forecasting in practice.

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